Design and Development of a Fuzzy Controller for Air Quality Assessment

Mamdani Fuzzy Inference System using Python (scikit-fuzzy)

Course: BE05000461 – Fuzzy Logic and Genetic Programming
Problem Based Learning (PBL)
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Abstract

Air quality is usually reported using sharp threshold bands, although human judgement of pollution is vague and several factors interact. This report presents a Mamdani fuzzy inference system that takes PM2.5, NO2 and wind speed as inputs and produces a single Air Quality Risk Index between 0 and 100. The controller uses trapezoidal and triangular membership functions, a base of 12 IF-THEN rules, min/max inference and centroid defuzzification. It was implemented with scikit-fuzzy and tested on 12 input combinations. The output rose smoothly with pollutant concentration and fell with increasing wind speed, as intended.

1. Problem

Standard air quality indices assign a category through fixed cut-offs. A reading of 59 µg/m³ PM2.5 may be "Good" while 61 µg/m³ is "Moderate", although the two are practically the same. Pollutants also interact: high PM2.5 together with high NO2 is more harmful than either alone, while wind disperses pollutants and reduces exposure. Experts describe these situations in linguistic terms such as "fairly high" or "calm day". Fuzzy logic can represent this reasoning directly, so it is a suitable tool for this problem.

2. Objectives

• Identify a real-world problem suited to fuzzy reasoning and define its inputs and output.

• Define linguistic variables and membership functions.

• Design a Mamdani rule base of at least 8 rules (12 used).

• Implement the system in Python using scikit-fuzzy.

• Test with at least 10 input combinations (12 used) and analyse the behaviour.

3. System Design

3.1 Inputs and output

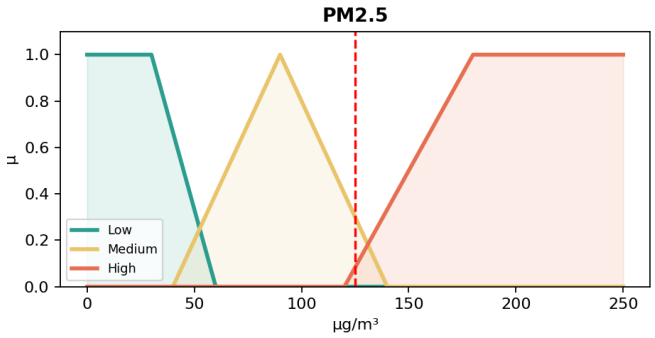
Variable	Role	Range	Linguistic terms
PM2.5	Input	0–250 µg/m³	Low, Medium, High
NO2	Input	0–200 µg/m³	Low, Medium, High
Wind speed	Input	0–20 km/h	Calm, Breezy, Windy
Risk index	Output	0–100	Good, Moderate, Unhealthy, Hazardous

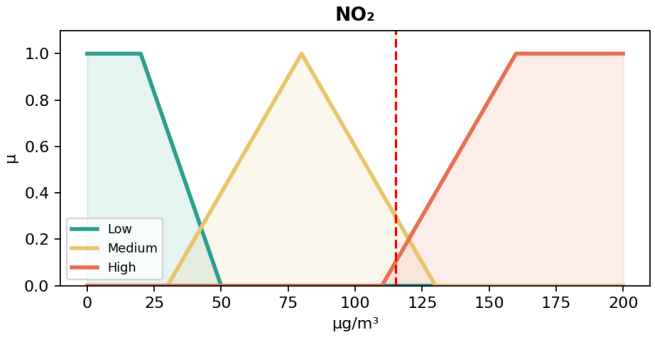
Higher PM2.5 and NO2 push the risk up, while higher wind speed pushes it down. Three terms per input give 27 possible input combinations, which keeps the rule base readable.

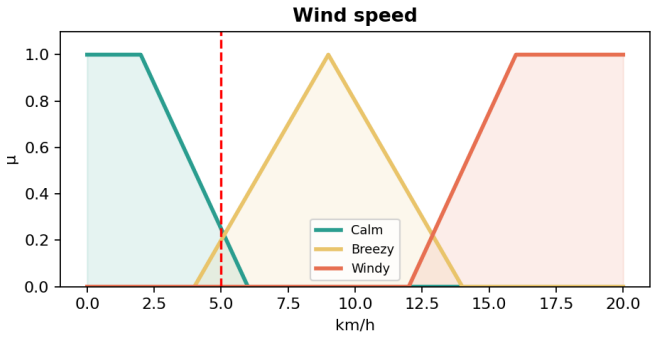
3.2 Membership functions

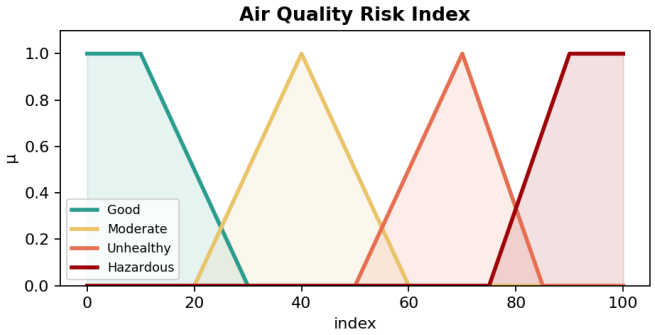
Trapezoidal functions are used at the ends of each range (so that "Low" and "High" saturate at 1) and triangular functions in the middle. A triangle is written as a trapezoid with b = c. Adjacent sets overlap intentionally to give smooth transitions.

Variable	Term	Shape	Parameters
PM2.5	Low	Trapezoid	0, 0, 30, 60
PM2.5	Medium	Triangle	40, 90, 140
PM2.5	High	Trapezoid	120, 180, 250, 250
NO2	Low	Trapezoid	0, 0, 20, 50
NO2	Medium	Triangle	30, 80, 130
NO2	High	Trapezoid	110, 160, 200, 200
Wind	Calm	Trapezoid	0, 0, 2, 6
Wind	Breezy	Triangle	4, 9, 14
Wind	Windy	Trapezoid	12, 16, 20, 20
Risk	Good	Trapezoid	0, 0, 10, 30
Risk	Moderate	Triangle	20, 40, 60
Risk	Unhealthy	Triangle	50, 70, 85
Risk	Hazardous	Trapezoid	75, 90, 100, 100









4. Rules

The rule base has 12 rules. AND is the minimum of memberships, and an empty antecedent means that input is ignored. Rules that point to the same output term are combined by max. A grid check over the whole input space confirmed that every input combination fires at least one rule.

#	PM2.5	NO2	Wind	THEN Risk
R1	High	High	—	Hazardous
R2	High	—	Calm	Hazardous
R3	High	—	—	Unhealthy
R4	Medium	High	—	Unhealthy
R5	Medium	Medium	Calm	Unhealthy
R6	Medium	Medium	—	Moderate
R7	Medium	Low	—	Moderate
R8	Low	Low	—	Good
R9	Low	Medium	—	Moderate
R10	Low	—	Windy	Good
R11	Low	High	—	Moderate
R12	—	High	Calm	Unhealthy

3.3 Worked example (hand calculation)

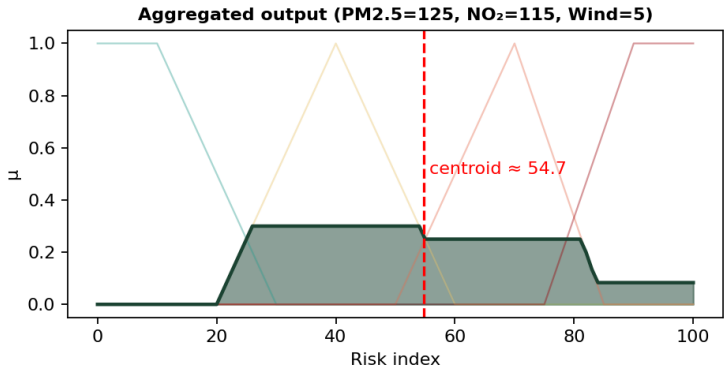
Sample reading: PM2.5 = 125 µg/m³, NO2 = 115 µg/m³, wind = 5 km/h. Using $\mu = (c - x)/(c - b)$ on the falling edges and $(x - a)/(b - a)$ on the rising edges:

Input	Term	Calculation	μ
PM2.5	Medium	(140–125)/50	0.300
PM2.5	High	(125–120)/60	0.083
NO2	Medium	(130–115)/50	0.300
NO2	High	(115–110)/50	0.100
Wind	Calm	(6–5)/4	0.250
Wind	Breezy	(5–4)/5	0.200

Rules that fire (all other terms have $\mu = 0$):

Rule	Firing strength w = min(...)	Output term
R1	min(0.083, 0.100) = 0.083	Hazardous
R2	min(0.083, 0.250) = 0.083	Hazardous
R3	0.083	Unhealthy
R4	min(0.300, 0.100) = 0.100	Unhealthy
R5	min(0.300, 0.300, 0.250) = 0.250	Unhealthy
R6	min(0.300, 0.300) = 0.300	Moderate
R12	min(0.100, 0.250) = 0.100	Unhealthy

Taking the strongest firing per term gives Moderate = 0.300, Unhealthy = 0.250, Hazardous = 0.083. With term peaks of 40, 70 and 95 the weighted average is (0.300×40 + 0.250×70 + 0.083×95) / 0.633 = 37.42 / 0.633 ≈ **59.1**. The centroid of the aggregated output set, which is what scikit-fuzzy returns, is ≈ **54.7**. Both fall in the Unhealthy band.



5. Implementation

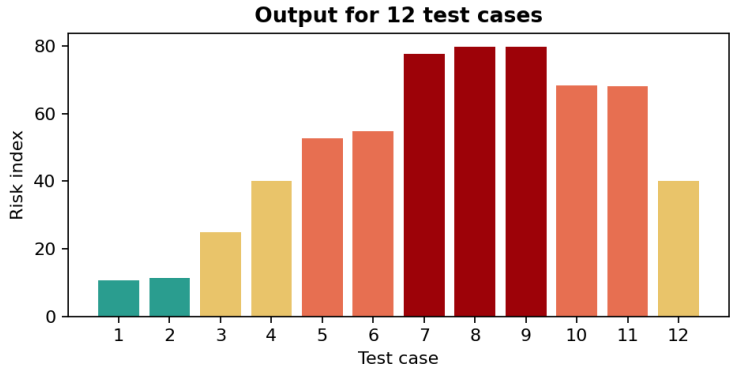
The controller is implemented in **air_quality_fuzzy_controller.py** using numpy, scikit-fuzzy and matplotlib. The steps are: (1) define Antecedent and Consequent objects with their universes; (2) assign membership functions with trapmf; (3) build the 12 rules from a rule table using the & operator for AND; (4) create a ControlSystem and a ControlSystemSimulation; (5) set inputs, call compute() and read the centroid output; (6) map the value to a category (below 25 Good, 25–50 Moderate, 50–75 Unhealthy, 75 and above Hazardous). The same file also contains the hand-calculation check, the 12 test cases and plotting functions.

6. Results

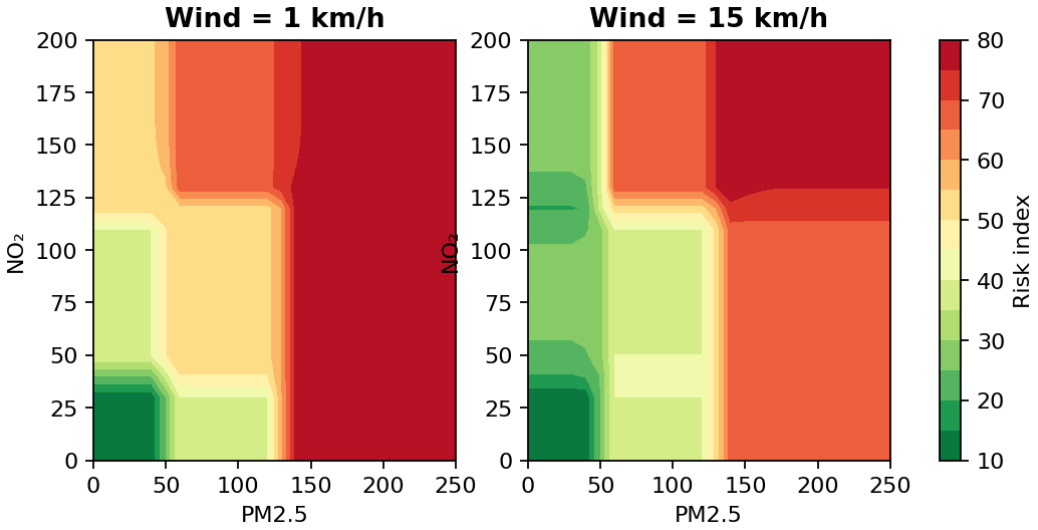
Twelve input combinations covering clean, moderate, polluted, calm and windy conditions were tested.

#	PM2.5	NO2	Wind	Risk index	Category
1	10	15	12	10.6	Good
2	25	30	15	11.4	Good
3	45	40	8	25.0	Moderate
4	70	60	10	40.0	Moderate
5	90	80	3	52.6	Unhealthy

#	PM2.5	NO2	Wind	Risk index	Category
6	125	115	5	54.7	Unhealthy
7	150	130	2	77.6	Hazardous
8	200	170	1	79.8	Hazardous
9	230	190	0.5	79.8	Hazardous
10	60	150	2	68.3	Unhealthy
11	160	40	18	68.2	Unhealthy
12	35	120	10	40.0	Moderate



The control surface below shows the risk index over PM2.5 and NO2 for calm wind (1 km/h) and windy conditions (15 km/h). The whole surface shifts toward lower risk when wind is high.



7. Analysis

Behaviour. The index increases monotonically with PM2.5 and NO2 and decreases with wind speed. Clean readings (cases 1–2) give about 11, while extreme readings (cases 7–9) saturate near 80. Intermediate readings give intermediate values, which a threshold system cannot do.

Strengths. Smooth graded output; readable rules that an expert can edit; natural handling of interacting pollutants and wind; full coverage of the input space.

Weaknesses. Centroid defuzzification cannot reach the ends of the range, so the index saturates near 80 instead of 100. Membership shapes and rules were set by judgement, not fitted to data. Only

three variables are used; O3, CO, SO2 and humidity are ignored. The results were not validated against measured data.

Scope for improvement. Add more inputs; tune membership parameters with a genetic algorithm; validate against real monitoring data; compare Sugeno inference and other defuzzifiers; rescale the output so that the maximum maps to 100.

8. Conclusion

A Mamdani fuzzy controller was designed and implemented to assess air quality from PM2.5, NO2 and wind speed. It converts vague, interacting conditions into one interpretable risk index without hard thresholds, and it behaved consistently on all 12 test cases.

References

[1] L. A. Zadeh, "Fuzzy Sets," Information and Control, vol. 8, no. 3, 1965.

[2] E. H. Mamdani and S. Assilian, "An Experiment in Linguistic Synthesis with a Fuzzy Logic Controller," Int. J. Man-Machine Studies, 1975.

[3] T. J. Ross, Fuzzy Logic with Engineering Applications, Wiley.

[4] scikit-fuzzy documentation, https://pythonhosted.org/scikit-fuzzy/